Confidence-Aware Temporal-First Graph Residual Refinement for Taxi Demand Forecasting

Tianqi Wang and Ligang He*

Department of Computer Science, University of Warwick, Coventry CV4 7AL, UK
tim.wang@warwick.ac.uk
ligang.he@warwick.ac.uk

Abstract. Hourly taxi demand forecasting must accommodate heterogeneous temporal patterns and uneven demand across urban zones. We investigate whether a separately trained residual stage can improve fixed zone-level temporal forecasts, and whether graph propagation and heuristic confidence provide benefits beyond node-wise correction. The framework combines cached temporal convolutional network (TCN) forecasts with residual learners over geographic or rolling origin–destination (OD) graphs. Across twelve 24-hour NYC evaluation windows and five residual-model seeds, a feature-matched residual MLP reduces MAE from 22.736 to 19.288. Geographic GraphSAGE obtains a lower mean MAE of 19.089, but its paired day-block interval relative to the MLP includes zero, so the independent contribution of message passing is not established. Enabling confidence as both a node feature and a loss weight reduces MAE from 19.475 to 19.228 in a matched 2×2 ablation. Scale-normalized error is not negatively associated with the confidence score, precluding a calibration interpretation. The results support modular residual refinement and a modest confidence contribution within the selected windows, without establishing superiority over end-to-end spatio-temporal graph models.

Keywords: Urban mobility analytics · Taxi demand forecasting · Graph neural networks · Residual learning · Confidence estimation · Big data analytics

1 Introduction

Urban mobility demand forecasting is a representative data-intensive analytics problem in modern transportation systems. Taxi platforms, dispatching systems, and mobility operators need fine-grained short-term demand estimates to support vehicle allocation, local supply balancing, and operational planning. In dense metropolitan areas, however, demand is highly heterogeneous across both time and space. Individual taxi zones may follow different activity rhythms due to commuting, nightlife, airports, commercial districts, or residential patterns, while the same zone may exhibit short-term fluctuations, daily cycles, weekly regularity, and longer-range seasonal changes. These characteristics make hourly

* Corresponding author.

zone-level taxi demand forecasting more difficult than a standard univariate time-series prediction task.

Recent work on traffic and mobility forecasting has increasingly used deep temporal models and graph neural networks to capture such spatio-temporal structure [7, 5]. Temporal models are effective at learning local sequential patterns, whereas graph neural networks can propagate information across related locations [12]. Nevertheless, directly using a graph model to predict demand from scratch can entangle two different modelling problems: learning each zone's temporal demand dynamics and learning spatially structured corrections across zones. This coupling may reduce interpretability and make it harder to identify whether improvements come from stronger temporal modelling or from spatial dependency learning. Decoupled spatio-temporal designs have therefore become attractive because they can separate inherent temporal signals from spatial diffusion or correction effects [10].

This paper studies a temporal-first residual correction pipeline rather than introducing a new temporal or graph operator. A temporal forecasting model first produces a base prediction for each taxi zone from historical pickup counts. A separately trained GraphSAGE module predicts corrections to these fixed forecasts, rather than predicting demand directly. This division gives the second-stage target an explicit interpretation as temporal forecast error. Rolling origin-destination (OD) graphs and a geographic graph provide alternative neighbourhoods for correction. The temporal-only reference has neither graph propagation nor a residual predictor, so it is not a matched control for message passing.

The second question concerns prediction reliability across zones with different demand support and temporal stability. Inspired by uncertainty-aware forecasting [1, 14], we combine historical demand support, predictive stability, and local temporal consistency into a bounded score. We call this score confidence, but it is a heuristic reliability indicator rather than a calibrated probability or guaranteed error estimate. Fixed, validation-selected, and learned component weights are evaluated as distinct variants. Its use as a residual feature is separate from its use to weight the training loss.

The framework addresses rolling citywide mobility analytics through reusable temporal forecasts and a separate residual stage. Revised feature, graph, and confidence experiments use the same cached TCN forecasts, graph-compatible cohort, all 288 target hours, and no residual shrinkage in their primary results. Legacy weight-selection and reliability analyses are reported separately because their shrinkage and coverage differ.

The main contributions of this work are summarized as follows.

- We formulate a modular taxi-demand correction pipeline in which GraphSAGE learns historical errors of a fixed temporal predictor. The contribution is the task-specific formulation and empirical study, not residual learning or temporal-first ordering alone.
- We specify a multi-source reliability score and distinguish its roles as a node feature, a training weight, and an error-analysis variable.

– We report matched residual controls, five-seed graph and confidence ablations, peak-correction sensitivity, completed-trip OD graphs, and residual-stage process scaling. These controls separate the robust residual benefit from the smaller and less certain graph-specific contribution.

2 Related Work

Urban mobility demand forecasting is commonly formulated at station, grid, road-segment, or zone level. Deep recurrent, convolutional, and attention-based models have become standard for nonlinear temporal forecasting [7]. In taxi and ride-hailing settings, multi-zone prediction remains difficult because zones differ in density, periodicity, and volatility [13, 14]. Prior ride-hailing studies have therefore used graph-based models to capture non-Euclidean relationships among urban regions [2]. Temporal models can capture local sequential patterns, but they do not explicitly model how prediction errors may co-vary across nearby or functionally connected zones. This motivates our temporal-first formulation: the temporal model is responsible for zone-specific demand dynamics, while the graph module is reserved for the remaining spatially structured residuals.

Graph neural networks model sensors or urban regions as nodes connected by spatial or functional edges. Spatio-temporal graph convolutional models show the value of graph message passing for traffic prediction [12], and surveys identify graph construction and dynamic dependency learning as core design choices [5]. OD flows, geographic adjacency, and learned dependencies can all encode useful cross-zone relationships [9]; OD-specific ride-sourcing models further show that graph representations can capture dependencies among origin-destination pairs [6]. However, many spatio-temporal GNNs couple temporal encoding and graph propagation in a single predictor, making it harder to separate improvements from stronger temporal modelling and improvements from graph-based correction. Decoupled designs such as D2STGNN address this issue by separating spatial and temporal components [10]. Our method follows the same motivation but applies it differently: GraphSAGE [3] is used after temporal prediction as a residual correction layer, not as the primary demand predictor.

Residual learning models corrections around a reference mapping rather than reconstructing the complete target function [4]. Here, the correction target is $y_{i,t} - \hat{y}_{i,t}^{(T)}$ after the temporal forecast has been produced. This is a system-level correction stage, not simply an internal skip connection. Both residual learning and decoupled temporal–spatial modelling are established ideas; we do not claim priority for either. Our formulation keeps temporal forecasts fixed while fitting a graph correction module, and studies reliability features and weighting in that setting. Comparing graph variants characterizes their observed performance, but the temporal-only reference cannot isolate the benefit of neighbourhood aggregation.

Reliability is also important in demand forecasting. Sparse zones and unstable periods can make point forecasts less trustworthy, motivating uncertainty-aware travel demand models [14]. Monte Carlo Dropout provides a practical predictive-

variance estimate [1]. These methods mainly focus on quantifying predictive uncertainty or probabilistic demand. Our confidence score has a narrower role: it is not a calibrated probability, but a bounded reliability signal derived from historical support, prediction stability, and local temporal consistency. The score is used as a feature for graph residual refinement, evaluated as an optional sample-weighting signal, and validated by comparing confidence with actual absolute error.

3 Problem Definition

Let $\mathcal{V}$ be the preprocessed taxi-zone set and $y_{i,t}$ the pickup count in zone i during hour t. Given historical pickup counts strictly before t, the task is to predict $\mathbf{y}_t$ for all evaluated zones. A temporal model outputs $\hat{y}_{i,t}^{(T)}$, and the graph module predicts a raw residual correction $\Delta\hat{y}_{i,t}$. Historical residual labels are $r_{i,t} = y_{i,t} - \hat{y}_{i,t}^{(T)}$. An experiment-specific multiplier determines how much of the predicted correction is added to the temporal forecast.

For a target hour, residual training uses only earlier labelled snapshots; the current target-hour features are reserved for inference and evaluation. Graphs may be geographic or OD-based, while the temporal-only reference omits correction. Revised OD graphs require both a pre-target pickup and a valid drop-off strictly before the target hour, so the destination is available at prediction time. Results are reported over rolling windows using MAE and RMSE, with repeated residual-model seeds where specified.

4 Method

4.1 Overview

Figure 1 separates target-hour inputs from historical residual supervision. A temporal predictor produces $\hat{y}_{i,t}^{(T)}$, and confidence estimation produces $c_{i,t}$ without observing $y_{i,t}$. A separately fitted GraphSAGE module predicts a raw correction $\Delta\hat{y}_{i,t}$ over a selected zone graph:

$$\hat{y}_{i,t} = \hat{y}_{i,t}^{(T)} + s_t \Delta\hat{y}_{i,t}. \tag{1}$$

The residual multiplier s_t is one in the primary revised experiments. A fixed grid applied only to saved peak-hour corrections is reported as a post-hoc sensitivity analysis, without selecting a test-optimal primary coefficient. Historical labels $r_{i,s} = y_{i,s} - \hat{y}_{i,s}^{(T)}$ for $s < t$ supervise the correction module. Target-hour labels are reserved for evaluation, and residual fitting does not update the temporal predictor.

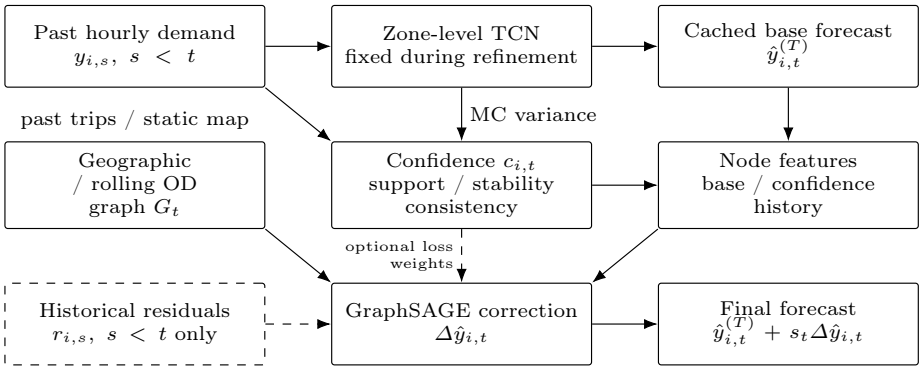


Fig. 1. Target-hour correction uses fixed temporal forecasts, graph structure, and node features. Dashed arrows denote training-only supervision or optional loss weighting; $y_{i,t}$ is not a training input. Historical means use past demand. Revised OD graphs include only trips completed before the target hour.

4.2 Temporal-First Demand Forecasting

For each zone, trip records are converted into a dense hourly pickup series, with missing hours filled by zero. The selected backbone for Ex2–Ex6 is the pure TCN implementation, not an LSTM/Transformer/GRU multi-branch model. Its reference configuration accepts the preceding 720 hourly counts. Four residual causal-convolution blocks use 64 channels, kernel size three, and dilations $1, 2, 4, 8$. Each block contains two convolutions with ReLU and dropout; the final timestep feeds a linear one-hour prediction head. With historical context $\mathcal{H}_{i,t}$, the base forecast is

$$\hat{y}_{i,t}^{(T)} = f_{\phi_i}(\mathcal{H}_{i,t}), \tag{2}$$

where ϕ_i is the checkpointed zone model. The reference training configuration uses historical MinMax scaling, Adam, learning rate 10^{-3}, weight decay 10^{-4}, and dropout 0.2. It allows 50 epochs, with training-loss early stopping after five non-improving epochs. At inference, 20 MC Dropout passes provide a mean forecast and variance in the original count scale. The cached residual runs reuse the base forecasts and confidence components rather than regenerating MC predictions. LSTM, GRU, Transformer, and other temporal models appear only in the backbone comparison. These numerical settings describe the reference implementation; saved checkpoint metadata records the training cutoff, not a complete configuration manifest.

4.3 Graph Residual Refinement

Each taxi zone is a graph node. The feature ablation includes base-only and base-plus-confidence inputs. The latter is

$$\mathbf{x}_{i,t} = [\hat{y}_{i,t}^{(T)}, c_{i,t}], \tag{3}$$

where $c_{i,t}$ is the confidence score. Historical means are concatenated in the combined feature variant and the cached graph and confidence-weighting runs:

$$\mathbf{x}_{i,t}^{+} = [\hat{y}_{i,t}^{(T)}, c_{i,t}, \bar{y}_{i,t}^{24}, \bar{y}_{i,t}^{168}, \bar{y}_{i,t}^{720}]. \tag{4}$$

The history-only feature variant omits $c_{i,t}$ from this vector. All confidence-weighting variants retain the same combined feature vector, so uniform loss weighting does not remove confidence from the inputs.

The reference residual module uses two 256-dimensional GraphSAGE layers, GELU, dropout 0.1, and a scalar linear head. It supports ordinary mean aggregation and edge-weighted mean aggregation. For a directed edge $j \to i$ with weight a_{ji}, the latter computes

$$\mathbf{m}_i = \frac{\sum_{j \to i} a_{ji} \mathbf{h}_j}{\max(\sum_{j \to i} |a_{ji}|, 10^{-12})}, \tag{5}$$

followed by separate self and neighbour linear maps. Both layers use the same aggregation choice. Historical snapshots are batched as disconnected graphs, with labels restricted to training locations. The reference optimizer is Adam with learning rate 0.01 and cosine annealing over at most 300 epochs. Unweighted validation SmoothL1 loss selects the checkpoint, with patience 40. Each target-hour residual fit starts from a fresh initialization; absent usable history, the temporal forecast is returned unchanged.

4.4 Multi-Source Confidence Estimation

Confidence combines three heuristic indicators. Define the clipping operation and historical support as

$$\mathrm{clip}(z) = \min(1, \max(0.05, z)), \qquad n_{i,t} = \sum_{s<t} y_{i,s}. \tag{6}$$

With $\ell_{i,t} = \log(1 + n_{i,t})$, the support component is

$$c_{i,t}^{\mathrm{prior}} = \mathrm{clip}\left(0.2 + 0.8 \frac{\ell_{i,t} - \ell_{\min,t}}{\ell_{\max,t} - \ell_{\min,t}}\right). \tag{7}$$

The minimum and maximum are computed across historical zone counts. When they coincide, the normalized term is set to one. Thus support is relative to the zone cohort, not an absolute probability of correctness.

Let $v_{i,t}$ be the MC predictive variance and $q_t = \max(\mathrm{median}_i\, v_{i,t}, 10^{-3})$, using finite variances. Stability and history consistency are

$$c_{i,t}^{\mathrm{stab}} = \mathrm{clip}\big(\exp[-v_{i,t}/(3q_t)]\big), \tag{8}$$

$$c_{i,t}^{\mathrm{hist}} = \mathrm{clip}\big(\exp[-\,\mathrm{std}_h\{\log(1 + \bar{y}_{i,t}^{h})\}/\tau]\big), \tag{9}$$

where $h \in \{24, 168, 720\}$ and $\tau = 1$. Each mean includes zero-filled hours strictly before t, and the standard deviation uses the population convention. Log transformation reduces sensitivity to demand magnitude, while exponential decay

maps dispersion to a bounded score. These transformations are design choices, not theoretical calibration guarantees. Missing support uses 0.4; missing variance uses 0.2, or one if no finite variance is available. Fewer than two valid history means gives a consistency score of 0.5.

Cached runs retain available component scores from the temporal cache. Their loader substitutes stability one when that component is missing and recomputes missing historical consistency. These fallback rules differ from fresh MC computation and should not be interpreted as additional uncertainty measurements. The full score is a weighted geometric combination:

$$c_{i,t} = \exp\left(\alpha_1 \log c_{i,t}^{\text{prior}} + \alpha_2 \log c_{i,t}^{\text{stab}} + \alpha_3 \log c_{i,t}^{\text{hist}}\right), \tag{10}$$

where $\alpha_j \geq 0$ and $\sum_j \alpha_j = 1$. Geometric fusion penalizes a weak component without letting a strong component cancel it additively. This is a modelling rationale, not a proof that the fusion predicts error. The implementation defaults to $(0.3, 0.4, 0.3)$; the equal-fixed sensitivity variant instead uses $(1/3, 1/3, 1/3)$. Grid and random search select weights using validation residual loss. Learned-softmax variants optimize component logits jointly with GraphSAGE. Feature scores and loss weights are distinct: changing loss-weight coefficients does not automatically change the cached confidence feature.

4.5 Confidence-Weighted Residual Learning

The residual training objective uses SmoothL1 loss with optional confidence weights:

$$\mathcal{L} = \frac{1}{|\mathcal{S}_{\text{train}}|} \sum_{(i,t)\in\mathcal{S}_{\text{train}}} \bar{w}_{i,t} \cdot \text{SmoothL1}\left(\Delta\hat{y}_{i,t}, y_{i,t} - \hat{y}_{i,t}^{(T)}\right), \tag{11}$$

where $\bar{w}_{i,t}$ is mean-normalized within the training batch. Ablations set $w_{i,t}$ to one, a single confidence component, the fixed full confidence score, or the learned-softmax score. These variants test whether confidence should influence residual training, but the core reliability claim is assessed separately through confidence-error analysis.

4.6 Graph Construction Strategies

The graph study compares the temporal-only reference with a feature-matched residual MLP, an edge-free GraphSAGE control, geographic GraphSAGE, and rolling OD GraphSAGE. The residual MLP removes graph operations while approximately matching the active GraphSAGE parameter budget. The edge-free control preserves the GraphSAGE self transformations but removes all cross-node edges. The geographic variant reads the fixed 258×258 symmetric matrix `edge_weight_matrix_geo.csv` archived with the original implementation. Every nonzero entry is converted to an edge. The graph comparison uses unweighted neighbourhood aggregation, so the stored matrix magnitudes are not

used. No matrix-generation script is preserved; this experiment therefore evaluates an archived fixed topology and should not be read as a newly reproducible polygon-contiguity construction. The OD60 variant counts trips with pickup timestamps in $[t-60 \text{ days}, t)$ and valid drop-off timestamps before t. It removes same-zone trips and excluded identifiers, requires at least one trip per edge, and retains the ten largest destination flows per origin. The graph comparison uses uniform residual loss so that loss weighting does not vary across controls.

Computational cost. Let N denote zones, E_t retained edges, H_t historical residual snapshots, d hidden width, and R_t optimization epochs. Given precomputed summaries and variance, confidence fusion costs $O(N)$; this excludes temporal training and MC inference. Sparse top-k retention bounds E_t by kN, but the current OD implementation also materializes and writes a dense $N \times N$ matrix. That step incurs quadratic storage and output cost despite sparse message passing. Residual training costs approximately $O(R_t H_t(E_t d + N d^2))$ for a fixed two-layer architecture; inference costs $O(E_t d + N d^2)$. Section 5.10 measures process scaling for independent residual fits, while explicitly excluding temporal computation and OD construction; the result is not an end-to-end complexity measurement.

5 Experiments

5.1 Dataset and Preprocessing

We evaluate the framework on 2021 NYC Taxi and Limousine Commission (TLC) trip records [8], aggregated into hourly pickup demand by taxi zone. Each trip contributes one pickup event to its pickup zone and pickup hour. The resulting target is a dense zone-hour matrix of pickup counts. Residual graph preprocessing excludes identifiers 46, 103, 104, 105, 264, and 265, as specified by the scripts. Further graph-name matching determines usable nodes, so the residual cohorts need not equal the all-zone temporal comparison. Experiment-specific coverage is reported below rather than assuming identical cohorts.

Pickup and drop-off timestamps are parsed without timezone localization or conversion; the stored Parquet schema contains timezone-naive timestamps. Window boundaries and peak-hour labels therefore use the source clock representation, interpreted as recorded New York local clock time rather than UTC. Daylight-saving ambiguities are not explicitly resolved by preprocessing.

The temporal input and historical demand summaries use pickup counts strictly before the target hour. Summaries cover 24, 168, and 720 hours. In the revised rolling OD evaluation, an eligible trip must have a pickup timestamp in $[t - L \text{ days}, t)$ and a valid drop-off timestamp strictly before t. Thus, its destination is available at the one-hour-ahead prediction instant. Same-zone trips and excluded identifiers are removed before retaining the ten largest destination flows per origin. We retain the earlier pickup-only construction only as a legacy definition and do not use it for the revised matched comparison.

5.2 Experimental Setup

The experiments follow a rolling one-hour-ahead protocol. The cached-temporal runs use twelve 24-hour windows, resulting in 288 target hours. Ex1 evaluates 12 windows $\times$ 24 hours $\times$ 259 zones, giving 74,592 zone-hour predictions for each temporal backbone. The revised residual experiments use one frozen cache of these TCN predictions, a fixed graph-compatible cohort of 258 locations, and a fixed 60/20/20 location split with 51 test locations. Each reported variant therefore has 14,688 test predictions per model seed. For each target hour, the residual learner is trained only on labelled snapshots earlier than that hour; when the required history partition is empty, the unchanged temporal forecast is returned and retained in the matched evaluation.

The twelve windows begin at 00:00 on March 1 and 29, April 26, May 24, June 21, July 5 and 19, August 16, September 13, October 11, November 8, and December 6, 2021. Each covers the following 24 hourly targets. These dates are the deterministic defaults of the original evaluation script and are retained unchanged for comparability across backbones and residual variants; they were neither sampled randomly nor selected using test error. No external event-stratification criterion was recorded. The windows should therefore be viewed as a fixed convenience schedule spanning March–December, not as a representative or continuous-time sample of 2021. Training, validation, and test partitions are location-based, not month-separated. Residual labels are temporally truncated within each rolling window; this is not strict month-to-month generalization with disjoint training and test months.

For residual-training partitioning, peak target hours are $\{6, 7, 8, 9, 16, 17, 18\}$ in the dataset's clock representation. Historical residual snapshots are split into peak and non-peak partitions, and each target uses only earlier snapshots from its partition. History resets at each window start. The revised Ex2, Ex3, and Ex5 primary results apply the complete predicted residual, $s_t = 1$, at every hour. This removes the hand-set peak multiplier from the main comparison. Ex4 and Ex6 are retained as legacy sensitivity and reliability analyses with $s_t = 0.3$ at peak hours; their absolute errors are therefore not placed on the revised matched leaderboard.

Performance is reported using MAE and RMSE, with MAE as the primary metric. Ex3 and Ex5 are repeated with residual-model seeds $\{11, 23, 42, 67, 89\}$ while holding the TCN cache and location split fixed. We report the mean and standard deviation across these five residual initializations. Paired uncertainty intervals resample the twelve evaluation days as blocks and keep the five observed seeds fixed; they quantify variation across the selected days, not the full uncertainty of retraining the temporal backbone or sampling new periods. Ex2 remains a seed-42 feature ablation. The temporal-backbone comparison still lacks a matched end-to-end spatio-temporal graph baseline.

The Ex identifiers refer to experiment scripts, not subsection order. Revised Ex2, Ex3, and Ex5 include all $12 \times 24 \times 51 = 14,688$ test predictions per variant and seed, including temporal fallbacks before a compatible residual-history partition becomes available. No finite prediction is removed in the matched Ex3

Table 1. Experiment identifiers and evaluation protocols. Revised Ex2, Ex3, and Ex5 use the same 51 test locations, all 288 target hours, and no residual shrinkage.

ID	Section	Main variable	Evaluation scope	Peak s_t
Ex1	5.3	Temporal model	74,592 zone-hours	–
Ex2	5.4	Node features	14,688, seed 42	1.0
Ex5	5.5	Residual/graph control	$5 \times 14{,}688$	1.0
Ex5-S	5.6	Peak multiplier sensitivity	$5 \times 14{,}688$	grid
Ex3	5.7	Feature × loss weighting	$5 \times 14{,}688$	1.0
Ex4	5.8	Weight selection (legacy)	14,076, seed 42	0.3
Ex6	5.9	Confidence group (legacy)	14,688, seed 42	0.3
Ex8	5.10	Residual CPU workers	8 fits/config.	–

or Ex5 analysis. Across five seeds, each Ex3/Ex5 variant therefore contributes 73,440 predictions. Ex1 retains its original all-zone scope, while Ex4 and Ex6 retain their explicitly labelled legacy protocols.

5.3 Temporal Backbone Selection

Table 2. Temporal backbone comparison over twelve rolling 24-hour windows. Lower values are better.

Temporal model	MAE	RMSE	MSE
SARIMA	73.32	145.53	21177.90
DLinear	31.04	54.86	3010.00
GRU	27.62	45.46	2066.90
LSTM	26.79	45.37	2058.20
PatchTST	26.38	45.18	2041.15
Transformer	21.80	37.47	1404.37
TCN	20.19	33.49	1121.56

Table 2 motivates the TCN temporal backbone used by the cached residual-refinement experiments. DLinear and PatchTST are reported from the dedicated rerun summary, while the other temporal baselines are reported from the overall temporal comparison. The residual experiments below use cached TCN predictions and location-based residual evaluation splits, so their absolute temporal-baseline MAE is not identical to the all-zone temporal-backbone summary in Ex1.

5.4 Residual Refinement Features

Table 3 isolates the input features by using the same uniform loss. Historical summaries provide the main feature-related gain: adding them to the base input reduces MAE from 20.178 to 19.404. Confidence alone does not improve MAE in this seed (20.203 versus 20.178), and adding confidence to the history variant

Table 3. Residual feature configurations in revised Ex2. All variants use uniform residual loss, seed 42, all 288 target hours, and $s_t = 1$. Lower values are better.

Method	MAE	RMSE	MAE improvement
Temporal baseline	22.736	33.321	–
Base feature residual GNN	20.178	29.867	11.25%
Base + confidence feature residual GNN	20.203	29.841	11.14%
Base + history feature residual GNN	19.404	28.529	14.66%
Base + confidence + history residual GNN	19.395	28.655	14.70%

changes MAE by only -0.009 while increasing RMSE from 28.529 to 28.655. Thus, Ex2 supports the value of historical demand summaries but does not by itself establish a confidence-feature benefit or seed robustness.

5.5 Graph Structure Comparison

Table 4. Matched residual and graph controls over five residual-model seeds. Values are mean ± standard deviation across seeds on the same 14,688 test points per seed. OD60 uses only trips completed before prediction time.

Model	MAE	RMSE	ΔMAE vs. MLP
Temporal TCN	22.736 ± 0.000	33.321 ± 0.000	+3.449
Residual MLP	19.288 ± 0.044	28.436 ± 0.069	0.000
GraphSAGE, no cross-node edges	19.251 ± 0.045	28.388 ± 0.049	−0.037
Geographic GraphSAGE	19.089 ± 0.047	28.207 ± 0.050	−0.199
OD60 GraphSAGE	19.391 ± 0.044	28.642 ± 0.074	+0.103

All four residual learners reduce MAE relative to the fixed TCN reference. The feature-matched residual MLP reduces MAE by 3.449 (15.2%), showing that most of the improvement can be obtained without cross-node aggregation. Geographic GraphSAGE has the lowest mean MAE in all five initializations, but its mean advantage over the residual MLP is 0.199; the paired day-block 95% interval for the MAE difference is $[-0.450, 0.021]$. Its advantage over edge-free GraphSAGE is 0.162 with interval $[-0.392, 0.038]$. Both intervals include zero. OD60 does not improve on the MLP control. These results establish residual-refinement benefit but do not isolate a statistically robust benefit from graph message passing over the selected twelve days.

5.6 Peak Residual Sensitivity

Table 5 shows that damping the saved peak-hour corrections lowers aggregate MAE for every residual learner. The originally used value 0.3 is close to, but

Table 5. MAE sensitivity to the multiplier applied only at peak hours. The same saved raw predictions are reused without refitting; entries are five-seed means. The grid is descriptive and is not used to select a test-optimal primary result.

Residual model	$s_t = 0$	$s_t = 0.3$	$s_t = 0.5$	$s_t = 0.7$	$s_t = 1$
Residual MLP	19.373	19.074	19.013	19.054	19.288
GraphSAGE, no cross-node edges	19.323	19.029	18.970	19.014	19.251
Geographic GraphSAGE	18.811	18.508	18.505	18.652	19.089
OD60 GraphSAGE	19.175	18.890	18.883	19.004	19.391

not consistently better than, 0.5; the difference for geographic GraphSAGE is only 0.003 MAE. Because all coefficients are evaluated on the same test targets, this sensitivity analysis does not justify selecting 0.5 as a new tuned setting. We retain $s_t = 1$ as the main no-heuristic result and report the grid to expose the effect of manual damping. A coefficient selected on separate validation periods or a learned gate would be required for a tuned primary claim.

5.7 Confidence Ablation

Table 6. Matched 2×2 confidence ablation over five residual-model seeds. Confidence is independently enabled as a node feature and as a fixed full-confidence loss weight. Values are mean ± standard deviation across seeds.

Confidence feature	Confidence weighting	MAE	RMSE	ΔMAE
Off	Off	19.475 ± 0.071	28.643 ± 0.123	0.000
On	Off	19.391 ± 0.044	28.642 ± 0.074	-0.085
Off	On	19.324 ± 0.050	28.510 ± 0.033	-0.152
On	On	19.228 ± 0.075	28.399 ± 0.102	-0.247

Relative to disabling both roles, the confidence feature alone reduces mean MAE by 0.085, fixed confidence weighting alone by 0.152, and their combination by 0.247 (1.27%). Paired day-block 95% intervals are $[-0.170, -0.008]$, $[-0.242, -0.063]$, and $[-0.355, -0.143]$, respectively. The combined setting is lower than the reference in every reported initialization. The effect is nevertheless small relative to the residual-refinement gain, and the intervals condition on the fixed TCN cache and five observed residual seeds. This ablation supports a modest empirical contribution from confidence under the evaluated protocol, not calibration or a theoretical reliability guarantee.

5.8 Confidence Weight Sensitivity

Table 7 reports the lowest MAE for random search among these legacy configurations. Its descriptive reduction relative to the equal-fixed rerun is approximately

Table 7. Legacy confidence component-weight sensitivity with peak multiplier 0.3. Coefficients are mean selected weights across evaluated targets. These values are not directly comparable to the revised Ex3 table.

Mode	MAE	RMSE	w_p	w_s	w_h
Fixed	17.74	31.70	0.33	0.33	0.33
Grid search	17.68	31.63	0.41	0.26	0.32
Random search	17.67	31.64	0.36	0.30	0.34
Learned softmax	17.72	31.93	0.18	0.64	0.18

0.40%. Because Ex4 uses one residual seed, excludes the first step, and applies peak shrinkage, it does not supersede the matched Ex3 result. Component-weight tuning remains a secondary sensitivity analysis rather than an established source of accuracy gain.

5.9 Reliability Analysis

Table 8. Legacy confidence-grouped reliability analysis on all Ex6 test predictions with peak multiplier 0.3.

Confidence group	Mean confidence	MAE	RMSE
Top 25%	0.928	11.63	19.91
Middle 50%	0.843	14.82	25.79
Bottom 25%	0.498	30.94	48.88

Table 8 reports lower absolute error in higher-confidence groups. Across all Ex6 test predictions, Pearson and Spearman correlations between confidence and absolute error are -0.314 and -0.278, respectively. After dividing absolute error by the pre-target 168-hour mean demand (lower-bounded by one), the corresponding correlations are approximately 0.063 and 0.060. Thus, the absolute-error association is strongly affected by demand scale. The grouped result is descriptive rather than a calibration measure or evidence that confidence independently predicts normalized error.

Table 9 directly compares the composite score with its components and simple support proxies without training another model. Stability is more strongly associated with absolute error than the composite score, whereas history consistency is near zero. Historical mean and prior support correlate positively with absolute error because higher-demand zones have larger count errors. After scale normalization, neither stability nor combined confidence retains the expected negative association. The combined score is therefore not superior to these simple comparators as an error-ranking statistic under this protocol.

Table 9. No-training reliability comparators on the same 14,688 Ex6 predictions. Entries are Spearman ρ with 95% intervals from resampling the twelve evaluation days as blocks. For stability, consistency, and combined confidence, a reliable score would ideally correlate negatively with error; historical mean and prior support are included as simple demand-scale comparators.

Signal	Absolute error	Scale-normalized error
Historical mean demand	0.617 [0.605, 0.629]	−0.225 [−0.237, −0.217]
Prior support	0.608 [0.597, 0.618]	−0.230 [−0.240, −0.222]
Prediction stability	−0.590 [−0.602, −0.578]	0.202 [0.190, 0.212]
History consistency	−0.022 [−0.052, 0.007]	−0.014 [−0.032, 0.004]
Combined confidence	−0.278 [−0.306, −0.253]	0.060 [0.047, 0.075]

5.10 Residual-Stage Runtime Scaling

Table 10. CPU process scaling for eight independent residual fits on one frozen real-data snapshot. Wall time includes worker startup. Aggregate RSS is sampled across parent and child processes and may count shared pages more than once.

Workers	Wall time (s)	Fits/s	Speedup	Peak RSS (GB)
1	97.8	0.0818	1.00	1.15
2	53.4	0.1499	1.83	1.68
4	31.2	0.2566	3.14	2.75

The benchmark uses width 256, 100 epochs, one CPU thread per worker, and eight independent fits with identical frozen inputs. Preparing the snapshot takes 4.58 s. Increasing the worker count from one to four reduces fit wall time from 97.8 to 31.2 s, corresponding to 3.14× process-level speedup, while sampled aggregate RSS increases from 1.15 to 2.75 GB. This benchmark demonstrates parallelism only for independent residual fits. It excludes TCN training and inference, rolling OD construction, data preparation outside the frozen snapshot, and end-to-end orchestration; it therefore does not establish whole-pipeline scalability.

6 Discussion

The matched controls distinguish the benefit of residual prediction from that of graph propagation. The residual MLP recovers most of the improvement over the fixed TCN, supporting the modular correction stage without requiring cross-node aggregation. Geographic GraphSAGE has the lowest mean MAE in all five initializations, but its paired day-block interval relative to both the MLP and edge-free GraphSAGE includes zero. The evidence therefore supports residual refinement more strongly than a graph-specific mechanism. OD60 does not improve on the MLP under the stricter completed-trip construction.

Confidence as a feature and confidence as a loss weight produce distinct, modest effects. In the five-seed matched ablation, enabling both lowers mean MAE by 0.247 (1.27%) relative to disabling both. The component roles are therefore empirically useful under this protocol, but much smaller than the residual-stage gain. In the no-training comparison, stability alone has a stronger absolute-error rank association than the composite score, while neither retains the expected sign after normalizing by historical demand scale. Consequently, the combined score remains a heuristic training and analysis signal rather than a calibrated or demonstrably superior reliability estimate.

Peak-hour sensitivity exposes a second explanation for apparently improved residual results. Multipliers around 0.3–0.5 reduce aggregate MAE for every residual learner, but 0.3 is not consistently optimal and the grid is evaluated on test targets. We therefore retain the unshrunk output as the primary result rather than treating manual damping as a learned confidence gate. Separate validation periods would be required to tune this coefficient credibly.

The evaluation uses one city, one taxi data source, and twelve non-contiguous daily windows. It does not establish cross-city transfer, continuous-period robustness, or strict out-of-month generalization. Five residual seeds quantify initialization variability conditional on a fixed TCN cache and fixed location split; they do not capture temporal-backbone retraining uncertainty. Ex2, Ex4, and Ex6 remain single-seed analyses. Direct end-to-end spatio-temporal graph results are also absent from the current evidence, so the study does not establish state-of-the-art accuracy or a general advantage of temporal-first ordering.

From a processing perspective, independent residual fits show a 3.14× wall-time speedup from one to four CPU workers while sampled aggregate RSS rises from 1.15 to 2.75 GB. This supports residual-stage process parallelism, not whole-pipeline scalability: the benchmark excludes TCN work, OD construction, and complete orchestration. Dense OD matrix materialization remains a bottleneck despite sparse message passing.

7 Conclusion

This paper presents a confidence-aware temporal-first graph residual refinement framework for hourly taxi demand forecasting. The framework first learns zone-specific temporal demand forecasts and then applies GraphSAGE to correct residual errors over graph structures built from historical information. Multi-source confidence combines historical prior, prediction stability, and local temporal consistency, and is used both as a reliability feature and, in selected variants, as a training weight.

Using fixed TCN forecasts, the revised NYC experiments show a stable reduction from temporal prediction to node-wise residual refinement. Geographic GraphSAGE gives the lowest mean error, but its small advantage over matched non-graph controls is not robust across the selected day blocks. Confidence features and weighting provide a smaller additional reduction, while normalized-error analysis does not support a calibration interpretation. Completed-trip OD

construction, peak-multiplier sensitivity, repeated residual seeds, and process-level runtime measurements make these boundaries explicit. Direct end-to-end spatio-temporal baselines, broader temporal coverage, and cross-city evaluation remain necessary before claiming architectural superiority or general applicability.

Data and Code Availability

The underlying 2021 trip records and taxi-zone resources are publicly available from the NYC TLC Trip Record Data portal [8]. The processed Parquet file contains trip-level pickup and drop-off timestamps and zone identifiers; the paper reports only aggregate errors and does not redistribute individual trip records. The forecasting implementation and the archived geographic matrix are available in the project repository [11]. The matrix-generation script is not preserved, as stated in Sect. 4.6; the fixed file is provided to make the evaluated topology inspectable. The public repository does not include the frozen TCN predictions or derived revision tables. Exact replay of the reported residual-stage analyses therefore requires regeneration of the temporal predictions.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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